

HCSM-06

Sector Coarse Correspondence

The Gauge Algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$
from the D_4 Endomorphism Algebra of the 14-Mode Multiplet
The Hodge Complex Standard Model

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Abstract

The Hodge Complex Standard Model assigns the 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus to one Standard Model generation. This paper derives the gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ as a Lie subalgebra of the D_4 commutant on the complexified 14-mode multiplet, through the endomorphism algebra of the D_4 isotypic multiplicity spaces. We prove:

- (i) the D_4 commutant on $\mathbb{C}^{14}$ decomposes as $\bigoplus_{\rho} M_{m_{\rho}}(\mathbb{C})$ with complex dimension 27;
- (ii) $\mathfrak{su}(3)$ embeds in the A_1 -block endomorphism algebra $M_3(\mathbb{C})$ as the traceless anti Hermitian subalgebra;
- (iii) **the A_1 block is uniquely selected as the physical color algebra by a dimension and invariance theorem: among the two three-dimensional multiplicity spaces M_{A_1} and M_E , only M_{A_1} is D_4 -invariant on its isotypic subspace, so only M_{A_1} can host the physical color triplet;**
- (iv) $\mathfrak{su}(2)$ embeds in the B_1 -block endomorphism algebra $M_2(\mathbb{C})$;
- (v) **the choice of B_1 over B_2 is a substrate-level labeling convention, forced by the outer automorphism of D_4 and by the D_4 -equivariance of all HCSM-native operators;**
- (vi) the five D_4 isotypic projectors span an abelian $\mathfrak{u}(1)^5$ subalgebra;
- (vii) the generators of these three factors close to $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$, a 16-dimensional real Lie subalgebra of the commutant.

On the full 56-dimensional Hodge sector at $\lambda = 4$, we derive the sector-coarse correspondence: the space of D_4 -invariant, sector-block-diagonal, multiplicity-scalar operators is exactly 15-dimensional, spanned by the three sector operators $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ on each of the five isotypic blocks. The identification of the A_1 factor with physical color is now a theorem; the identification of the B_1 factor with physical weak isospin is a labeling convention with no physical consequences. All algebraic claims are verified at machine precision.

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1 Introduction

1.1 The gauge algebra problem

The Hodge Complex Standard Model (HCSM) identifies the 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation [4].

On this multiplet the D_4 point group of the square acts by permutation, and the paper HCSM-05 [5] established the D_4 isotypic decomposition.

The present paper asks: what Lie algebra emerges from the D_4 isotypic structure of this multiplet? The answer is the gauge algebra

$$\mathfrak{g}_{\text{HCSM}} = \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5, \quad (1)$$

realized as a Lie subalgebra of the D_4 commutant on the complexified 14-mode multiplet $\mathbb{C}^{14}$.

1.2 The one-line summary

The 14-mode multiplet at $\lambda = 4$ decomposes under D_4 as $3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E$. The D_4 commutant on this multiplet is the endomorphism algebra $\bigoplus_{\rho} M_{m_{\rho}}(\mathbb{C})$, of complex dimension 27. The gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ embeds in the commutant: $\mathfrak{su}(3)$ acts on the A_1 multiplicity space, $\mathfrak{su}(2)$ on the B_1 multiplicity space, and the five isotypic projectors span the $\mathfrak{u}(1)^5$ center. The A_1 identification with physical color is forced by dimension matching; the B_1 identification with physical weak isospin is a labeling convention fixed by the outer automorphism of D_4 .

1.3 Main results

- (1) D_4 isotypic decomposition (Theorem 3.1). The 14-mode multiplet decomposes as

$$V_{14} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E, \quad (2)$$

of complex dimension 14.

- (2) Commutant dimension (Theorem 4.3). By Schur's lemma, the D_4 commutant has complex dimension

$$\dim_{\mathbb{C}} \text{Comm}(D_4, V_{14}) = \sum_{\rho} m_{\rho}^2 = 3^2 + 1^2 + 2^2 + 2^2 + 3^2 = 27. \quad (3)$$

- (3) Endomorphism algebra (Theorem 4.4). The commutant is

$$\text{Comm}(D_4, V_{14}) \cong \bigoplus_{\rho \in \{A_1, A_2, B_1, B_2, E\}} M_{m_{\rho}}(\mathbb{C}), \quad (4)$$

of complex dimension 27.

- (4) $\mathfrak{su}(3)$ subalgebra (Theorem 5.3). The traceless anti-Hermitian subalgebra of $M_{m_{A_1}}(\mathbb{C}) = M_3(\mathbb{C})$ is a copy of $\mathfrak{su}(3)$ embedded in the commutant.
- (5) A_1 **uniqueness theorem (Theorem 5.4)**. Among the two three-dimensional multiplicity spaces M_{A_1} and M_E , only M_{A_1} admits an $\mathfrak{su}(3)$ action on a D_4 -trivial three-dimensional subspace. Therefore physical color $\mathfrak{su}(3)_c$ is the $\mathfrak{su}(3)$ subalgebra on M_{A_1} .
- (6) $\mathfrak{su}(2)$ subalgebra (Theorem 5.7). The traceless anti-Hermitian subalgebra of $M_{m_{B_1}}(\mathbb{C}) = M_2(\mathbb{C})$ is a copy of $\mathfrak{su}(2)$ embedded in the commutant.
- (7) B_1/B_2 **outer automorphism (Theorem 5.9)**. The outer automorphism $\varphi : D_4 \rightarrow D_4$ defined by $\varphi(r) = r$, $\varphi(s) = rs$ swaps the two irreps $B_1 \leftrightarrow B_2$ while fixing A_1, A_2, E . No D_4 -equivariant operator on V_{14} implements φ . Therefore the choice of B_1 (rather than B_2) as physical weak isospin is a substrate-level labeling convention.

- (8) $\mathfrak{u}(1)^5$ subalgebra (Theorem 5.12). The five isotypic projectors $P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E$ span an abelian Lie subalgebra isomorphic to $\mathfrak{u}(1)^5$.
- (9) Closure (Theorem 5.13). The generators of the three factors above close to a Lie subalgebra of $\text{Comm}(D_4, V_{14})$ isomorphic to $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$, of real dimension $8 + 3 + 5 = 16$.
- (10) Sector-coarse correspondence (Theorem 6.3). On the full 56-dimensional Hodge sector at $\lambda = 4$, the space of D_4 -invariant, sector-block-diagonal, multiplicity-scalar operators is exactly 15-dimensional, spanned by the three sector operators $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ on each of the five isotypic blocks.

1.4 Disclosed inputs

The framework's substrate-level disclosed inputs, inherited unchanged from HCSM-01 [2], are:

- (S1) The substrate is the periodic square lattice $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ with the D_4 point group of the square as its symmetry group.
- (S2) The scalar Laplacian is the 5-point Laplacian, the unique nearest-neighbour, D_4 -invariant, positive-semidefinite operator with kernel equal to the constant functions.
- (S3) The substrate size $L = 8$ is selected by equal entropy spacing (HCSM-01 [2]) and by maximal cancellation of the winding generator (HCSM-14.b [9]).

Two further inputs are used in the present paper:

- (P1) *Identification thesis*. The 14 modes at $\lambda = 4$ are in one-to-one correspondence with the 14 Standard Model particles of one generation (HCSM-23 [10]).
- (C1) *Structural premise of color*. Physical color is a D_4 -invariant internal index, i.e., it transforms trivially under the substrate's point group. This is the same structural statement the Standard Model makes: color is an internal gauge index, orthogonal to spacetime.

No other inputs are used.

1.5 Cross references

Depends on: HCSM-00 [1], HCSM-01 [2], HCSM-02 [3], HCSM-04 [4], HCSM-05 [5], HCSM-07 [6], HCSM-09 [8].

Propagates to: HCSM-08 [7], HCSM-38 [11], HCSM-39 [12], HCSM-40 [13], HCSM-42 [14], HCSM-44 [15].

2 Setup

2.1 The substrate and the mode multiplet

The substrate is the periodic square lattice

$$\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8, \quad N = 64, \tag{5}$$

consisting of $N = 64$ sites arranged in an 8×8 grid with periodic boundary conditions [2]. The discrete 5-point Laplacian has eigenvalue

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (n_1, n_2) \in \{0, \dots, 7\}^2. \quad (6)$$

At the self-paired eigenvalue $\lambda = 4$, the mode set M has 14 elements [4]:

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad (7)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad (8)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \quad (9)$$

We work with the complexified mode space $V_{14} = \mathbb{C}^{14}$ with basis $\{e_m : m \in M\}$. The physical mode vectors on the torus are real; the complexification is the natural home of the gauge algebra, as in quantum field theory.

2.2 The D_4 action

The D_4 point group of the square acts on Λ by

$$r : (n_1, n_2) \mapsto (-n_2, n_1) \pmod{8}, \quad s : (n_1, n_2) \mapsto (n_1, -n_2) \pmod{8}, \quad (10)$$

subject to $r^4 = s^2 = (rs)^2 = e$. The permutation action on the 14 modes extends to a representation

$$\rho_{14} : D_4 \rightarrow GL(V_{14}), \quad \rho_{14}(g)e_m = e_{g \cdot m}. \quad (11)$$

We denote by $R = \rho_{14}(r)$ and $S = \rho_{14}(s)$ the two generators.

2.3 The D_4 character table

The D_4 irreps are five: two 1-dimensional real representations A_1, A_2 , two 1-dimensional real representations B_1, B_2 , and one 2-dimensional real representation E . We use the standard conjugacy classes $\{e\}, \{r^2\}, \{r, r^3\}, \{s, r^2s\}, \{rs, r^3s\}$ with sizes 1, 1, 2, 2, 2:

irrep ρ	e	r^2	r, r^3	s, r^2s	rs, r^3s
A_1	1	1	1	1	1
A_2	1	1	1	-1	-1
B_1	1	1	-1	1	-1
B_2	1	1	-1	-1	1
E	2	-2	0	0	0

Table 1: D_4 character table.

3 The D_4 isotypic decomposition

Theorem 3.1 (D_4 isotypic decomposition of V_{14}). *Under D_4 , the complexified 14-mode multiplet decomposes as*

$$V_{14} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E, \quad (12)$$

with multiplicities $(m_{A_1}, m_{A_2}, m_{B_1}, m_{B_2}, m_E) = (3, 1, 2, 2, 3)$.

Proof. Direct decomposition via the character formula. The character of the permutation module on the 14 modes is

$$\chi_{V_{14}} = (14, 2, 0, 2, 2), \quad (13)$$

on the conjugacy classes $\{e\}, \{r^2\}, \{r, r^3\}, \{s, r^2s\}, \{rs, r^3s\}$. Then

$$m_\rho = \frac{1}{8} \sum_C |C| \chi_{V_{14}}(C) \chi_\rho(C)^*, \quad (14)$$

giving $(3, 1, 2, 2, 3)$. Dimension check: $3 + 1 + 2 + 2 + 6 = 14$. $\square$

Remark 3.2 (Isotypic decomposition). *The isotypic decomposition of V_{14} is*

$$V_{14} = \bigoplus_{\rho} V_{\rho} \otimes M_{\rho}, \quad (15)$$

where V_{ρ} is the irreducible representation space of ρ and M_{ρ} is the multiplicity space of dimension m_{ρ} . Explicitly:

$$\dim V_{A_1} = 1, \dim M_{A_1} = 3; \quad \dim V_{A_2} = 1, \dim M_{A_2} = 1; \quad \dim V_{B_1} = 1, \dim M_{B_1} = 2; \quad (16)$$

$$\dim V_{B_2} = 1, \dim M_{B_2} = 2; \quad \dim V_E = 2, \dim M_E = 3. \quad (17)$$

Remark 3.3 (Isotypic dimensions). *The A_1 block has isotypic dimension $1 \times 3 = 3$. The E block has isotypic dimension $2 \times 3 = 6$. This difference— A_1 trivial in V_{ρ} , E non-trivial—is the structural fact that will select A_1 as the color algebra in Section 5.2.*

4 The D_4 commutant

4.1 Definition and Schur's lemma

Definition 4.1 (D_4 commutant). *The D_4 commutant on V_{14} is the algebra of D_4 -equivariant endomorphisms,*

$$\text{Comm}(D_4, V_{14}) = \{X \in \text{End}_{\mathbb{C}}(V_{14}) : [X, R] = [X, S] = 0\}. \quad (18)$$

Theorem 4.2 (Schur decomposition of the commutant). *For a D_4 -representation $V = \bigoplus_{\rho} V_{\rho} \otimes M_{\rho}$ with $\dim M_{\rho} = m_{\rho}$,*

$$\text{Comm}(D_4, V) \cong \bigoplus_{\rho} \text{End}_{\mathbb{C}}(M_{\rho}) = \bigoplus_{\rho} M_{m_{\rho}}(\mathbb{C}). \quad (19)$$

Proof. Standard Schur's lemma. Since the V_{ρ} are pairwise non-isomorphic, every D_4 -equivariant endomorphism preserves the isotypic decomposition; on each block $V_{\rho} \otimes M_{\rho}$, an equivariant endomorphism must act as $\text{id}_{V_{\rho}} \otimes A_{\rho}$ with $A_{\rho} \in \text{End}(M_{\rho})$. The general element is thus $\bigoplus_{\rho} \text{id}_{V_{\rho}} \otimes A_{\rho}$. $\square$

Theorem 4.3 (Commutant dimension). *The D_4 commutant on the complexified 14-mode multiplet has complex dimension*

$$\dim_{\mathbb{C}} \text{Comm}(D_4, V_{14}) = \sum_{\rho} m_{\rho}^2 = 3^2 + 1^2 + 2^2 + 2^2 + 3^2 = 27. \quad (20)$$

Proof. Immediate from Theorem 4.2 and the multiplicities $(3, 1, 2, 2, 3)$. $\square$

4.2 The endomorphism algebra

Theorem 4.4 (Endomorphism algebra structure). *The commutant decomposes explicitly as*

$$\text{Comm}(D_4, V_{14}) \cong M_3(\mathbb{C}) \oplus M_1(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus M_3(\mathbb{C}), \quad (21)$$

with $\mathbb{C}$ -dimensions $9 + 1 + 4 + 4 + 9 = 27$.

Proof. Immediate from Theorems 4.2 and 4.3. The five blocks correspond to the five irreps A_1, A_2, B_1, B_2, E in that order. $\square$

Remark 4.5 (Block decomposition and physical sectors). *The five matrix blocks are the natural Lie-algebraic factors:*

- $M_3(\mathbb{C})$ on the A_1 multiplicity space: complex dimension 9, real dimension 18.
- $M_1(\mathbb{C})$ on the A_2 multiplicity space: complex dimension 1, real dimension 2.
- $M_2(\mathbb{C})$ on the B_1 multiplicity space: complex dimension 4, real dimension 8.
- $M_2(\mathbb{C})$ on the B_2 multiplicity space: complex dimension 4, real dimension 8.
- $M_3(\mathbb{C})$ on the E multiplicity space: complex dimension 9, real dimension 18.

The gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ embeds in this algebra, with $\mathfrak{su}(3)$ in the A_1 block, $\mathfrak{su}(2)$ in the B_1 block, and $\mathfrak{u}(1)^5$ spanned by the identity elements of all five blocks.

5 The gauge algebra

5.1 The $\mathfrak{su}(3)$ subalgebra on the A_1 multiplicity space

The A_1 multiplicity space $M_{A_1} \cong \mathbb{C}^3$ is the home of the color algebra. Let $\{u_1, u_2, u_3\}$ be an orthonormal basis of M_{A_1} .

Definition 5.1 (Embedding map). *The embedding of the A_1 block endomorphism algebra is*

$$\phi_{A_1} : M_3(\mathbb{C}) \rightarrow \text{End}_{\mathbb{C}}(V_{14}), \quad \phi_{A_1}(X) = \sum_{a,b=1}^3 u_a X_{ab} u_b^\dagger, \quad (22)$$

extended by zero on the other isotypic blocks.

Proposition 5.2 (Homomorphism). *ϕ_{A_1} is a unital algebra homomorphism and a Lie algebra homomorphism:*

$$\phi_{A_1}(XY) = \phi_{A_1}(X)\phi_{A_1}(Y), \quad \phi_{A_1}([X, Y]) = [\phi_{A_1}(X), \phi_{A_1}(Y)]. \quad (23)$$

Proof. Direct computation. $\square$

Theorem 5.3 ($\mathfrak{su}(3)$ subalgebra). *The traceless anti-Hermitian subalgebra of $M_3(\mathbb{C})$,*

$$\mathfrak{su}(3) = \{X \in M_3(\mathbb{C}) : X^\dagger = -X, \text{tr}(X) = 0\}, \quad (24)$$

embeds in $\text{Comm}(D_4, V_{14})$ via ϕ_{A_1} as a real 8-dimensional compact Lie subalgebra.

Proof. The image $\phi_{A_1}(\mathfrak{su}(3))$ is contained in the A_1 block of the commutant, and ϕ_{A_1} preserves brackets by Proposition 5.2. The Lie algebra $\mathfrak{su}(3)$ is generated by the anti-Hermitian traceless 3×3 matrices, of real dimension 8. Its standard physics form uses the Gell-Mann matrices λ^a ($a = 1, \dots, 8$) with Hermitian generators $T^a = \lambda^a/2$ satisfying

$$[T^a, T^b] = i \sum_c f^{abc} T^c, \quad (25)$$

where f^{abc} are the $\mathfrak{su}(3)$ structure constants. The embedding is $T_c^a = \phi_{A_1}(\lambda^a/2)$, $a = 1, \dots, 8$. $\square$

5.2 The A_1 uniqueness theorem

The commutant $\text{Comm}(D_4, V_{14})$ contains *two* 3-dimensional matrix blocks: M_{A_1} and M_E . Both are candidates for hosting an $\mathfrak{su}(3)$ subalgebra. The following theorem selects the unique candidate compatible with physical color.

Theorem 5.4 (A_1 uniqueness for physical color). *Let M_{A_1} and M_E be the two 3-dimensional multiplicity spaces in $\text{Comm}(D_4, V_{14})$. Then:*

- (a) $V_{A_1} \otimes M_{A_1} = \mathbb{C}^3$ is a D_4 -trivial 3-dimensional subspace of V_{14} , i.e., every $g \in D_4$ acts as the identity on V_{A_1} .
- (b) $V_E \otimes M_E = \mathbb{C}^2 \otimes \mathbb{C}^3$ is a 6-dimensional subspace of V_{14} , decomposing under D_4 as $E \otimes \mathbb{C}^3$, i.e., a D_4 -doublet tensored with the multiplicity space.
- (c) Physical color is a 3-dimensional internal index that transforms trivially under the D_4 symmetry of the substrate: the structural premise (C1). Therefore physical color is a $\mathfrak{su}(3)$ subalgebra of the commutant acting on a 3-dimensional D_4 -trivial subspace.
- (d) The unique such subspace of V_{14} is $V_{A_1} \otimes M_{A_1}$.
- (e) Consequently, physical color $\mathfrak{su}(3)_c$ is the $\mathfrak{su}(3)$ subalgebra of the commutant on M_{A_1} .

Proof. (a) V_{A_1} is 1-dimensional and carries the trivial representation A_1 . Hence $V_{A_1} \otimes M_{A_1} = \mathbb{C}^3$ with D_4 acting trivially on V_{A_1} and trivially on the whole space by D_4 -equivariance of the isotypic decomposition.

(b) V_E is 2-dimensional and carries the irrep E . Hence $V_E \otimes M_E = \mathbb{C}^2 \otimes \mathbb{C}^3$ with D_4 acting as E on the first factor and trivially on the second. Dimension $= 2 \times 3 = 6$.

(c) This is premise (C1).

(d) From (a) and (b), the only 3-dimensional subspace of V_{14} on which D_4 acts trivially is $V_{A_1} \otimes M_{A_1}$. The candidates $V_E \otimes M_E$ (dimension 6, non-trivial D_4 action) and $V_{B_1} \otimes M_{B_1}$, $V_{B_2} \otimes M_{B_2}$ (dimension 2, non-trivial) fail.

(e) From (c) and (d), the $\mathfrak{su}(3)$ subalgebra of the commutant acting on the color triplet must be the one on M_{A_1} . $\square$

Remark 5.5 (Status of the color identification). *Theorem 5.4 replaces the earlier framework input “ $A_1 \leftrightarrow$ physical color” with a derived statement. The only retained premise is (C1), the D_4 -triviality of color, which is the natural statement that color is an internal gauge index.*

5.3 The $\mathfrak{su}(2)$ subalgebra on the B_1 multiplicity space

The B_1 multiplicity space $M_{B_1} \cong \mathbb{C}^2$ is the home of the weak-isospin algebra. Let $\{v_1, v_2\}$ be an orthonormal basis.

Definition 5.6 (Embedding map). $\phi_{B_1} : M_2(\mathbb{C}) \rightarrow \text{End}_{\mathbb{C}}(V_{14})$ is defined by

$$\phi_{B_1}(X) = \sum_{a,b=1}^2 v_a X_{ab} v_b^\dagger, \quad (26)$$

extended by zero on the other isotypic blocks.

Theorem 5.7 ($\mathfrak{su}(2)$ subalgebra). The traceless anti-Hermitian subalgebra $\mathfrak{su}(2) \subset M_2(\mathbb{C})$ embeds in $\text{Comm}(D_4, V_{14})$ via ϕ_{B_1} as a real 3-dimensional compact Lie subalgebra.

Proof. Same structure as Theorem 5.3. The Lie algebra $\mathfrak{su}(2)$ is generated by the anti-Hermitian traceless 2×2 matrices; the standard physics basis uses the Pauli matrices σ^i , with Hermitian generators $T^i = \sigma^i/2$ satisfying $[T^i, T^j] = i\epsilon^{ijk}T^k$. $\square$

5.4 The B_1/B_2 outer automorphism and the labeling convention

The commutant $\text{Comm}(D_4, V_{14})$ contains *two* 2-dimensional matrix blocks: M_{B_1} and M_{B_2} . Both support isomorphic $\mathfrak{su}(2)$ subalgebras. We now show that the choice between them is a labeling convention, forced by the outer automorphism of D_4 .

Theorem 5.8 (Outer automorphism of D_4). The map $\varphi : D_4 \rightarrow D_4$ defined by $\varphi(r) = r$, $\varphi(s) = rs$ is a group automorphism of D_4 . It is not inner (no conjugation by an element of D_4 realizes it), and it exchanges the two reflection conjugacy classes

$$\{s, r^2s\} \longleftrightarrow \{rs, r^3s\}, \quad (27)$$

thereby swapping $B_1 \leftrightarrow B_2$ while fixing A_1, A_2, E .

Proof. Check the relations: $\varphi(r)^4 = r^4 = e$; $\varphi(s)^2 = (rs)^2 = rsrs = r(r^3s)s = r^4s^2 = e$; $(\varphi(r)\varphi(s))^2 = (r \cdot rs)^2 = (r^2s)^2 = r^2sr^2s = r^2(r^2s)s = r^4s^2 = e$. So φ preserves the D_4 presentation. Its effect on the character table: on B_1 , $\chi_{B_1}(s) = +1$, $\chi_{B_1}(rs) = -1$; on B_2 , $\chi_{B_2}(s) = -1$, $\chi_{B_2}(rs) = +1$. Applying φ exchanges the two reflection classes, so $B_1 \leftrightarrow B_2$. $\square$

Theorem 5.9 (B_1/B_2 labeling convention). The outer automorphism φ of Theorem 5.8 acts on $\text{Comm}(D_4, V_{14})$ by exchanging the two $M_2(\mathbb{C})$ blocks $M_{B_1} \leftrightarrow M_{B_2}$. No D_4 -equivariant operator on V_{14} implements φ . Therefore the choice of B_1 (rather than B_2) as the physical weak-isospin block is a substrate-level labeling convention.

Proof. The outer automorphism φ acts on the representation category $\text{Rep}(D_4)$ by permuting irreps: $B_1 \leftrightarrow B_2$, with A_1, A_2, E fixed. Hence on the isotypic decomposition $V_{14} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E$, the automorphism exchanges the two B -blocks. On the commutant $\bigoplus_{\rho} M_{m_{\rho}}(\mathbb{C})$, it exchanges the corresponding $M_2(\mathbb{C})$ factors.

For the second statement: no D_4 -equivariant operator on V_{14} implements φ , because φ is not an element of D_4 . The set of D_4 -equivariant operators on V_{14} is exactly the commutant $\text{Comm}(D_4, V_{14})$, which is generated by $R = \rho_{14}(r)$ and $S = \rho_{14}(s)$, both of which commute with D_4 and hence cannot implement an outer automorphism. All HCSM-native operators (the processor T of HCSM-13, the winding generator H_{wind} of HCSM-14, the zero state s of HCSM-22, the sector operators $\mathbb{I}, \mathcal{F}, \mathcal{O}$ of this paper, the orientation flip σ of HCSM-09) are D_4 -equivariant, so none of them distinguishes B_1 from B_2 . $\square$

Corollary 5.10 (Physical content invariance). *The physical content of HCSM—gauge algebra, fermion representation content, anomaly cancellation (HCSM-08 [7]), hypercharge quantization (HCSM-07 [6]), sector-coarse correspondence (Section 6), and all downstream predictions—is invariant under the relabeling $B_1 \leftrightarrow B_2$.*

Proof. The gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ is invariant under φ because φ acts as the identity on $\mathfrak{su}(3)$ (from the A_1 block), permutes the two $\mathfrak{su}(2)$ factors (which are isomorphic), and permutes the $\mathfrak{u}(1)$ projectors associated with B_1 and B_2 . The fermion representation content is determined by the D_4 isotypic decomposition, which is invariant under φ . The anomaly cancellation conditions of HCSM-08 are weighted sums over the fermion content; φ permutes the B_1 and B_2 contributions, and since both carry the same $\mathfrak{su}(2)$ content and the same hypercharge magnitude multiset, the anomaly sums are invariant. The hypercharge quantization in $(1/6)\mathbb{Z}$ is invariant. The sector-coarse correspondence of Section 6 is invariant because the sector operators $\mathbb{I}, \mathcal{F}, \mathcal{O}$ do not distinguish B_1 from B_2 . $\square$

5.5 The $\mathfrak{u}(1)^5$ subalgebra from the isotypic projectors

Definition 5.11 (Isotypic projectors). *For each irrep ρ of D_4 , the isotypic projector is*

$$P_\rho = \frac{\dim \rho}{|D_4|} \sum_{g \in D_4} \chi_\rho(g)^* \rho_{14}(g). \quad (28)$$

The five projectors $P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E$ satisfy $P_\rho P_{\rho'} = \delta_{\rho\rho'} P_\rho$ and $\sum_\rho P_\rho = \mathbb{I}_{14}$.

Theorem 5.12 ($\mathfrak{u}(1)^5$ subalgebra). *The five isotypic projectors span an abelian Lie subalgebra of $\text{Comm}(D_4, V_{14})$ isomorphic to $\mathfrak{u}(1)^5$, of real dimension 5.*

Proof. Each P_ρ commutes with the D_4 action by construction, so $P_\rho \in \text{Comm}(D_4, V_{14})$. The projectors are mutually orthogonal ($P_\rho P_{\rho'} = 0$ for $\rho \neq \rho'$) and sum to $\mathbb{I}_{14}$, giving a 5-dimensional span. As a Lie algebra with the commutator bracket, this span is abelian (all brackets vanish), hence isomorphic to $\mathfrak{u}(1)^5$. $\square$

5.6 Closure of the gauge algebra

Theorem 5.13 (Closure). *The image*

$$\phi_{A_1}(\mathfrak{su}(3)) \oplus \phi_{B_1}(\mathfrak{su}(2)) \oplus \text{span}\{P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E\} \quad (29)$$

is a Lie subalgebra of $\text{Comm}(D_4, V_{14})$ isomorphic to $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$, of real dimension $8+3+5=16$.

Proof. We verify the brackets.

- (1) $\mathfrak{su}(3)$ – $\mathfrak{su}(3)$ brackets. ϕ_{A_1} is a Lie algebra homomorphism, so $[\phi_{A_1}(X), \phi_{A_1}(Y)] = \phi_{A_1}([X, Y]) \in \phi_{A_1}(\mathfrak{su}(3))$.
- (2) $\mathfrak{su}(2)$ – $\mathfrak{su}(2)$ brackets. Same as (1).
- (3) $\mathfrak{su}(3)$ – $\mathfrak{su}(2)$ brackets. $\phi_{A_1}(\mathfrak{su}(3))$ is supported only on the A_1 multiplicity space; $\phi_{B_1}(\mathfrak{su}(2))$ is supported only on the B_1 multiplicity space. The two are orthogonal in $\text{End}_{\mathbb{C}}(V_{14})$, so $[\phi_{A_1}(X), \phi_{B_1}(Y)] = 0$.

- (4) $\mathfrak{su}(3)$ – $\mathfrak{u}(1)$ brackets. Each projector P_ρ acts as the identity on the ρ -block and zero elsewhere. On the A_1 block, P_{A_1} acts as $\mathbb{I}_{M_{A_1}}$, which commutes with every element of $M_{A_1}(\mathbb{C})$ (in particular with $\phi_{A_1}(\mathfrak{su}(3))$). Similarly for $P_{A_2}, P_{B_1}, P_{B_2}, P_E$ on the A_1 block (they act as zero). Hence $[\phi_{A_1}(X), P_\rho] = 0$ for all $X \in \mathfrak{su}(3)$ and all ρ .
- (5) $\mathfrak{su}(2)$ – $\mathfrak{u}(1)$ brackets. Same as (4).
- (6) $\mathfrak{u}(1)$ – $\mathfrak{u}(1)$ brackets. The projectors are mutually commuting, so their span is abelian.

The algebra thus closes to $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$, a direct sum of the three factors, of real dimension $8 + 3 + 5 = 16$. $\square$

Remark 5.14 (Sub-algebra of the commutant). *The gauge algebra is a 16-dimensional real Lie subalgebra of the 27-dimensional complex commutant $\bigoplus_\rho M_{m_\rho}(\mathbb{C})$. The commutant has real dimension 54, so the gauge algebra is a proper subspace; the remaining directions correspond to the “extra” structure not used by the gauge algebra (specifically, the traceless parts of the A_2 , B_2 , and E multiplicity blocks).*

6 Sector-coarse correspondence

The derivation above was on the 14-dimensional visible sector $\Omega^0|_{\lambda=4}$. We now extend to the full 56-dimensional Hodge sector at $\lambda = 4$.

6.1 The full Hodge sector

The full 56-dimensional sector at $\lambda = 4$ is

$$V_{56} = \Omega^0|_{\lambda=4} \oplus \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}, \quad (30)$$

with D_4 decomposition (from HCSM-02 [3])

$$V_{56} = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E. \quad (31)$$

Dimension check: $9 + 5 + 7 + 7 + 28 = 56$.

6.2 Sector operators

Definition 6.1 (Sector operators). *On V_{56} , define the three sector operators*

$$\mathbb{I} = P_{\Omega^0} + P_{\Omega^1} + P_{\Omega^2}, \quad \mathcal{F} = P_{\Omega^0} - P_{\Omega^1} + P_{\Omega^2}, \quad \mathcal{O} = P_{\Omega^0} - P_{\Omega^2}. \quad (32)$$

Restricted to the three sectors $(\Omega^0, \Omega^1, \Omega^2)$, their signatures are

$$\mathbb{I}|_{\text{sectors}} = (1, 1, 1), \quad \mathcal{F}|_{\text{sectors}} = (1, -1, 1), \quad \mathcal{O}|_{\text{sectors}} = (1, 0, -1). \quad (33)$$

6.3 The sector-coarse sub-commutant

Definition 6.2 (Sector-coarse sub-commutant). *The sector-coarse sub-commutant is the space of operators $X \in \text{End}_{\mathbb{C}}(V_{56})$ satisfying:*

- (i) $[X, R] = [X, S] = 0$ (D_4 -invariance);

(ii) $X = P_{\Omega^0}XP_{\Omega^0} + P_{\Omega^1}XP_{\Omega^1} + P_{\Omega^2}XP_{\Omega^2}$ (sector block-diagonality);

(iii) X acts as a scalar on each (ρ, Ω^k) block, where $\rho \in \{A_1, A_2, B_1, B_2, E\}$ and $k \in \{0, 1, 2\}$.

Theorem 6.3 (Sector-coarse correspondence). *The sector-coarse sub-commutant has complex dimension exactly $15 = 5 \times 3$ (five irreps $\times$ three Hodge sectors). It is spanned by the three sector operators $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ acting on each isotypic block.*

Proof. By Schur's lemma applied to each (ρ, Ω^k) block, a D_4 -invariant, sector-block-diagonal, multiplicity-scalar operator is determined by 15 complex scalar coefficients—one for each of the $5 \times 3 = 15$ blocks. Hence the space has complex dimension 15.

To see that the span of $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ on each block has the same dimension: for each fixed ρ , the restrictions of $(\mathbb{I}, \mathcal{F}, \mathcal{O})$ to the three sectors give three vectors in $\mathbb{C}^3$:

$$\mathbb{I}^{(\rho)} = (1, 1, 1), \quad \mathcal{F}^{(\rho)} = (1, -1, 1), \quad \mathcal{O}^{(\rho)} = (1, 0, -1). \quad (34)$$

The matrix with rows $(\mathbb{I}^{(\rho)}, \mathcal{F}^{(\rho)}, \mathcal{O}^{(\rho)})$ is

$$M = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 0 & -1 \end{pmatrix}, \quad \det M = 1(1 - 0) - 1(-1 - 1) + 1(0 + 1) = 4 \neq 0. \quad (35)$$

So the three operators are linearly independent and span $\mathbb{C}^3$ on each ρ -block. Over the five irreps, the span has complex dimension $5 \times 3 = 15$. The two spaces coincide. $\square$

Corollary 6.4 (Orientation $\mathbb{Z}_2$). *A sector-coarse functional Y on V_{56} is a complex linear combination*

$$Y = \sum_{\rho} (\alpha_{\rho} \mathbb{I} + \beta_{\rho} \mathcal{F} + \gamma_{\rho} \mathcal{O}) P_{\rho}, \quad (36)$$

of the three sector operators on each isotypic block. Under the sign flip $\mathcal{O} \mapsto -\mathcal{O}$, the multiset of physical sector values is preserved, up to relabeling of Ω^0 and Ω^2 .

Proof. Immediate from Theorem 6.3 and the antisymmetry of $\mathcal{O}$ under the D_4 orientation flip $\Omega^0 \leftrightarrow \Omega^2$. The full orientation- $\mathbb{Z}_2$ theorem is the subject of HCSM-09 [8]. $\square$

7 Inputs and derived

7.1 Derived from the substrate

The following results are derived from the substrate and the D_4 action alone:

- (1) The D_4 isotypic decomposition $V_{14} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E$ (Theorem 3.1).
- (2) The complex dimension of the D_4 commutant, $\dim_{\mathbb{C}} \text{Comm}(D_4, V_{14}) = 27$ (Theorem 4.3).
- (3) The endomorphism algebra decomposition $\text{Comm}(D_4, V_{14}) \cong M_3(\mathbb{C}) \oplus M_1(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus M_3(\mathbb{C})$ (Theorem 4.4).
- (4) The embedding of $\mathfrak{su}(3)$ in the A_1 block (Theorem 5.3).
- (5) **The A_1 uniqueness for physical color (Theorem 5.4).**

- (6) The embedding of $\mathfrak{su}(2)$ in the B_1 block (Theorem 5.7).
- (7) **The B_1/B_2 outer automorphism and labeling convention (Theorems 5.8, 5.9).**
- (8) The embedding of $\mathfrak{u}(1)^5$ from the isotypic projectors (Theorem 5.12).
- (9) The closure of the gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ (Theorem 5.13).
- (10) The sector-coarse correspondence on the 56-dimensional Hodge sector: 15-dimensional sub-commutant (Theorem 6.3).

7.2 Inputs

- (S1)–(S3) Substrate, Laplacian, and $L = 8$ selection (HCSM-01 [2], HCSM-14.b [9]).
- (P1) The identification of the 14 modes with one Standard Model generation (HCSM-23 [10]).
- (C1) The structural premise that physical color is a D_4 -invariant internal index. This is the standard statement in quantum field theory that color is an internal gauge quantum number, orthogonal to spacetime.

7.3 What is not claimed

We do not claim to derive the Standard Model gauge group $SU(3)_c \times SU(2)_L \times U(1)_Y$ from first principles. We derive the Lie algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ as a subalgebra of the D_4 commutant. The reduction to the physical gauge algebra involves premise (C1) and the labeling convention for B_1 vs. B_2 .

The framework’s claim is: any physically realized gauge theory with Standard Model content and the D_4 structure of the 14-mode multiplet must be compatible with a subalgebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ of the D_4 commutant. The physical content of the gauge algebra is then selected by premise (C1) and by the labeling convention of Theorem 5.9.

8 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSM companion repository at `12-VERIFICATION/HCSM-06_verification`. It constructs the substrate Laplacian on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, computes the 14-mode multiplet and its D_4 permutation representation, and verifies:

- (1) The D_4 isotypic decomposition of V_{14} (Theorem 3.1).
- (2) The commutant dimension $\dim_{\mathbb{C}} \text{Comm}(D_4, V_{14}) = 27$ (Theorem 4.3).
- (3) The endomorphism algebra decomposition $\bigoplus_{\rho} M_{m_{\rho}}(\mathbb{C})$ (Theorem 4.4).
- (4) The $\mathfrak{su}(3)$ embedding in the A_1 block (Theorem 5.3).
- (5) **The isotypic dimension argument of Theorem 5.4:** $\dim(V_{A_1} \otimes M_{A_1}) = 3$ **with trivial D_4 action;** $\dim(V_E \otimes M_E) = 6$ **with nontrivial D_4 action.**
- (6) The $\mathfrak{su}(2)$ embedding in the B_1 block (Theorem 5.7).
- (7) **The B_1/B_2 outer automorphism (Theorems 5.8, 5.9).**

- (8) The $\mathfrak{u}(1)^5$ subalgebra from the isotypic projectors (Theorem 5.12).
- (9) The bracket closure of $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ (Theorem 5.13).
- (10) The 15-dimensional sector-coarse sub-commutant (Theorem 6.3).

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

9 Conclusion

This paper has established the sector-coarse correspondence of the Hodge Complex Standard Model: the D_4 isotypic sectors of the 14-mode multiplet at $\lambda = 4$ correspond to the gauge algebra structure through the endomorphism algebra of their multiplicity spaces.

The 14-mode multiplet decomposes under D_4 as $3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E$. The D_4 commutant on the complexified multiplet is the endomorphism algebra $\bigoplus_{\rho} M_{m_{\rho}}(\mathbb{C})$, of complex dimension 27. The gauge algebra

$$\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5 \tag{37}$$

embeds in the commutant: $\mathfrak{su}(3)$ acts on the A_1 multiplicity space (dim 3), $\mathfrak{su}(2)$ on the B_1 multiplicity space (dim 2), and the five isotypic projectors span the $\mathfrak{u}(1)^5$ center.

The derivation is HCSM-native: it uses only the D_4 action on the 14-mode multiplet at $\lambda = 4$ and the standard endomorphism algebra structure. No imported machinery (no path integrals, no SYK kernels, no gauge fields as fundamental objects) is used.

On the full 56-dimensional Hodge sector at $\lambda = 4$, the sector-coarse correspondence is that the space of D_4 -invariant, sector-block-diagonal, multiplicity-scalar operators is exactly 15-dimensional, spanned by the sector operators $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ on each of the five isotypic blocks. This 15-dimensional space is the natural home of the physical hypercharge functional, whose specific coefficients are determined by the Standard Model content and are the subject of HCSM-07.

The identification of the derived $\mathfrak{su}(3)$ factor with physical color is a theorem (Theorem 5.4): the A_1 block is the unique 3-dimensional D_4 -invariant subspace of V_{14} , so it is the unique candidate for physical color. The identification of the derived $\mathfrak{su}(2)$ factor with physical weak isospin is a labeling convention (Theorem 5.9): the two blocks B_1 and B_2 are exchanged by the outer automorphism of D_4 , and no D_4 -equivariant operator distinguishes them. The physical content of the framework is invariant under this relabeling.

The framework derives the algebraic structure that any physically realized gauge theory with the Standard Model content and the D_4 structure of the 14-mode multiplet must be compatible with.

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}
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